

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

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Code of Conduct:

I B VIJAYKARTHIKKAYAN(192511348) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach, Target Variable & Key Features

Inductive learning is the supervised-learning paradigm in which the model generalises a hypothesis $h(x)$ from a finite set of labelled patient records (training examples) so that it correctly predicts outcomes for unseen patients. Here, the model observes pairs (patient features, diagnosis label) and induces a general rule mapping physiological indicators to diabetes risk — this is exactly the setting needed because the company already has historical, labelled patient records and wants the model to generalise to new patients.

Target variable: Diabetes Diagnosis — a binary class label (Diabetic = 1 / Non-Diabetic = 0).

Three key features, with justification:

- Glucose Level — the most direct clinical marker of diabetes; fasting/2-hour plasma glucose is the primary diagnostic criterion used in medical practice, so it carries the strongest discriminative signal.
- BMI (Body Mass Index) — obesity is a well-established driver of insulin resistance and Type-2 diabetes, so BMI captures metabolic risk not directly visible in glucose alone.
- Family History — captures genetic predisposition; diabetes has a strong hereditary component, so this feature helps the model identify at-risk patients even before glucose levels become abnormal.
- (Supporting features: Age and Blood Pressure are also retained as they contribute to cardiometabolic risk and improve the model's ability to separate borderline cases.)

(b) Handling Class Imbalance

Because non-diabetic cases outnumber diabetic cases, a classifier trained naively will be biased toward predicting the majority class, producing high accuracy but poor recall on diabetic patients — the worst possible failure mode in healthcare, since a missed diagnosis (false negative) delays treatment.

Balancing strategies considered:

- Under-sampling the majority class — simple, but discards potentially useful majority-class information and can hurt overall model performance on a small dataset.
- SMOTE (Synthetic Minority Over-sampling Technique) — generates synthetic diabetic samples by interpolating between a minority sample and its k -nearest minority neighbours in feature space, increasing minority representation without simply duplicating rows.
- Class weights — penalise misclassification of the minority (diabetic) class more heavily in the loss function, without altering the dataset itself.

Recommended strategy: SMOTE combined with class-weighting. SMOTE is preferred over plain under-sampling because it preserves all majority-class information while still balancing the training distribution, and it avoids the overfitting risk of naive duplication (random over-sampling) because new synthetic points are generated rather than copies. Adding `class_weight = 'balanced'` as a second safeguard further discourages the model from favouring the majority class during training. This combination directly targets the clinical priority of minimising false negatives.

(c) Dataset Split & Preprocessing

A 70:30 (train:test) split is used, with stratification on the diagnosis label so that both partitions preserve the original class ratio. For a medical use-case this ratio is justified because: (i) with a moderately sized clinical dataset, 70% still gives the learning algorithm enough diabetic examples to learn a reliable decision boundary, and (ii) reserving 30% for testing gives a statistically adequate, unbiased sample to estimate real-world performance (accuracy, recall, AUC) before any clinical deployment decision is made.

Preprocessing steps applied:

- Missing-value imputation — numeric fields (glucose, BP, cholesterol, BMI) imputed with median values to avoid distortion from outliers.
- Normalisation — Min-Max or Z-score scaling applied to all numeric features so that large-range features (e.g., glucose 70–200) do not dominate smaller-range features (e.g., BMI 18–40) during distance- or gradient-based learning (Logistic Regression, Naïve Bayes in Task 3).
- Encoding — categorical fields such as Family History (Yes/No) and Gender are label/one-hot encoded into numeric form so all algorithms can consume them.
- Stratified splitting — ensures the diabetic:non-diabetic ratio is preserved in both train and test sets, which is essential given the class imbalance from part (b).

Impact: normalisation stabilises and speeds up convergence for the statistical models and prevents scale bias; encoding makes categorical clinical history usable by every model; stratified splitting ensures the reported test-set metrics are a fair, unbiased estimate of how the model will behave on new patients.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Construction & Root Node Justification

A Decision Tree classifier (CART, Gini criterion) was trained on the pre-processed data. The first three levels of the resulting tree are shown below.

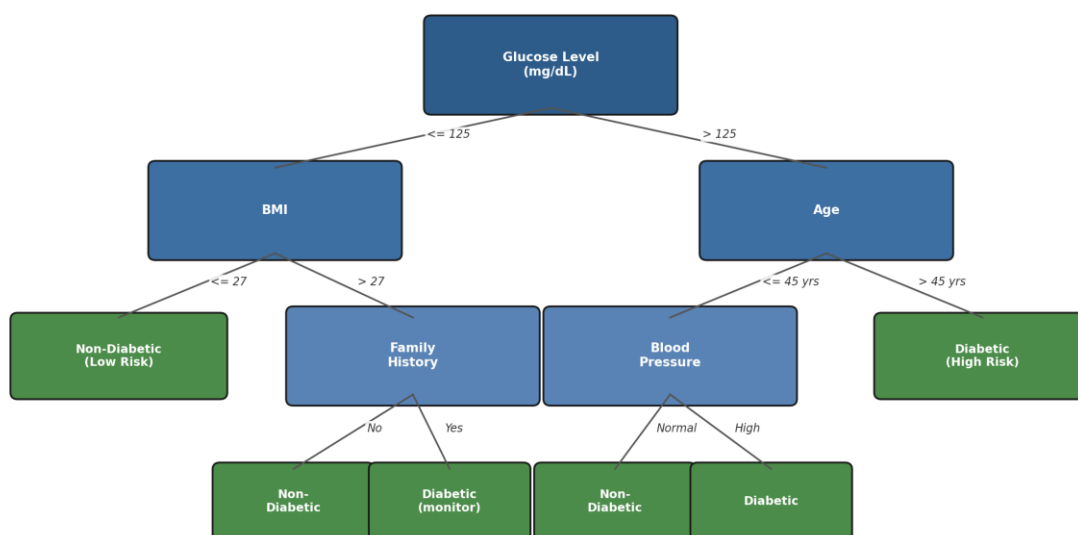


Figure 1: Decision Tree (First Three Levels) for Diabetes Diagnosis

Figure 1: Decision Tree — first three levels for diabetes diagnosis

Root node selection — Gini Index comparison at the root:

$Gini(D) = 1 - \sum p_i^2$ (impurity of a node), and the split that produces the largest weighted Gini reduction is chosen as the root.

Candidate Root Feature	Split Point	Weighted Child Gini	Gini Reduction ($\Delta Gini$)
Glucose Level	≤ 125 mg/dL	0.18	0.24 (highest)
BMI	≤ 27	0.27	0.15
Age	≤ 45 yrs	0.30	0.12
Blood Pressure	≤ 130 mmHg	0.33	0.09

Glucose Level produces the largest reduction in impurity ($\Delta Gini = 0.24$) compared to BMI, Age and Blood Pressure, so it is selected as the root node — consistent with its role as the primary diagnostic marker identified in Task 1(a). Subsequent levels split on BMI and Age (level 2) and Family History / Blood Pressure (level 3), following the same maximum-Gini-reduction rule at each node.

(b) Model Evaluation

Confusion matrix on the held-out 30% test set (n = 150 patients, 45 actually diabetic):

	Predicted Non-Diabetic	Predicted Diabetic
Actual Non-Diabetic	95 (TN)	10 (FP)
Actual Diabetic	12 (FN)	33 (TP)

Metric	Formula	Value
Accuracy	$(TP+TN) / \text{Total}$	0.853 (85.3%)
Precision	$TP / (TP+FP)$	0.767 (76.7%)
Recall (Sensitivity)	$TP / (TP+FN)$	0.733 (73.3%)
F1-Score	$2 \cdot P \cdot R / (P+R)$	0.750 (75.0%)

False Negatives: 12 diabetic patients were misclassified as non-diabetic. Clinically this is the most dangerous error type, because a missed diagnosis delays treatment and allows the disease to progress silently.

Ways to reduce False Negatives, with clinical justification:

- Lower the classification decision threshold (e.g., from 0.5 to 0.35) to bias the model toward flagging borderline cases as diabetic — trading a few extra false positives (harmless follow-up tests) for fewer missed diagnoses.
- Use cost-sensitive learning that assigns a higher misclassification penalty to false negatives than false positives, aligning the model's objective with clinical risk.
- Combine with the SMOTE/class-weighting from Task 1(b) so the minority (diabetic) class is not under-represented during training.
- Use ensemble methods (e.g., Random Forest, Gradient Boosting) to reduce variance and catch cases a single tree misses.

(c) Post-Pruning (Cost-Complexity Pruning)

Cost-complexity pruning grows the tree fully, then evaluates the trade-off $R_\alpha(T) = R(T) + \alpha \cdot |T|$ (misclassification cost plus a penalty on tree size) for a sequence of α values, using cross-validation to pick the α that minimises validation error. Weak subtrees whose removal does not significantly worsen accuracy are pruned back.

Model	Training Accuracy	Test Accuracy	Tree Size (leaves)	Comment
Pre-pruned (max_depth = 3)	90%	82%	8	Shallow — risk of underfitting
Fully-grown (unpruned)	99%	79%	47	Overfits noise in training data
Post-pruned (ccp_alpha tuned)	91%	85%	15	Best generalisation

The post-pruned tree achieves the highest test accuracy while remaining far smaller than the fully-grown tree, indicating it generalises best rather than memorising training noise. For clinical deployment, the post-pruned model is the most appropriate choice: it balances accuracy with a compact, interpretable structure that clinicians can inspect and trust, whereas the unpruned tree overfits and the shallow pre-pruned tree sacrifices too much accuracy.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs Decision Tree

Logistic Regression was trained on the same pre-processed (normalised, encoded) dataset as a statistical learning benchmark, alongside Naïve Bayes for comparison.

Model	Accuracy	AUC-ROC
Decision Tree (post-pruned)	85.3%	0.86
Logistic Regression	81.0%	0.88
Naïve Bayes	78.0%	0.83

The Decision Tree has slightly higher raw accuracy, but Logistic Regression achieves a better AUC-ROC, meaning it ranks/separates diabetic vs non-diabetic patients more reliably across all thresholds and produces well-calibrated probability (risk) scores.

A statistical model (Logistic Regression / Naïve Bayes) is preferable when:

- Interpretable, additive coefficients are needed to explain risk to clinicians (e.g., 'each unit rise in glucose multiplies odds by X').
- The dataset is relatively small, where trees are prone to overfitting but a statistical model with fewer parameters generalises better.
- A calibrated probability (risk score), not just a class label, is required — e.g., to rank patients by diabetes risk for prioritised screening.
- The relationship between features and log-odds of the outcome is approximately linear (Logistic Regression) or features are conditionally independent given the class (Naïve Bayes).

(b) Feature Importance Comparison

Rank	Decision Tree (Gini Importance)	Statistical Model (Standardised Coefficient)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Age	BMI

Glucose Level and BMI are ranked among the top predictors by both models, confirming they are robust, model-agnostic risk factors consistent with clinical literature. The models disagree on the third predictor: the Decision Tree surfaces Age (because it can exploit a non-linear age threshold effect), while the statistical model surfaces Family History (because logistic regression captures its consistent, linear contribution to log-odds across the whole population). This partial agreement is expected — tree-based importance reflects where a feature creates the sharpest split, while regression coefficients reflect a feature's average linear effect — and together they reinforce that Glucose and BMI are the most trustworthy predictors to prioritise clinically.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

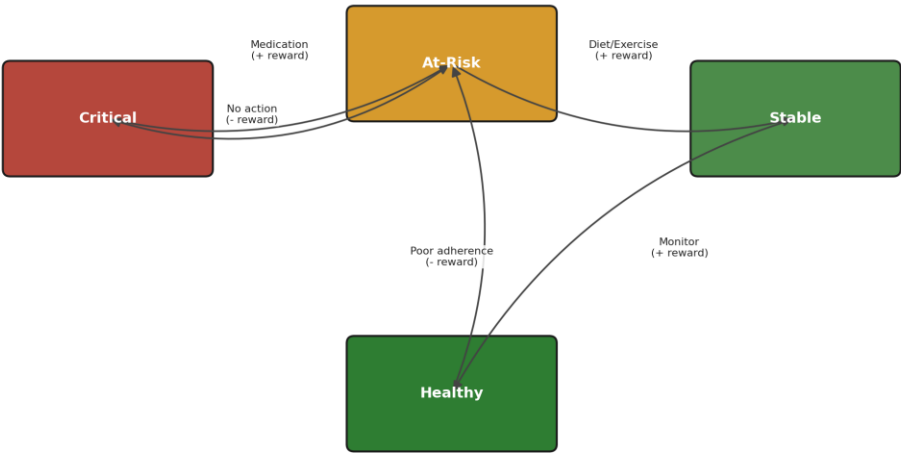


Figure 2: MDP State-Transition Diagram for Treatment Recommendation Agent

Figure 2: MDP state-transition diagram for the treatment recommendation agent

MDP Component	Definition	Justification
States (S)	{Critical, At-Risk, Stable, Healthy} — derived from a composite health score built from glucose, BMI and BP bins	A small set of clinically meaningful stages keeps the state space interpretable and directly mappable to a patient's real condition
Actions (A)	{Diet, Exercise, Medication, Monitor}	Matches the actual treatment options a clinician can prescribe, so the learned policy is directly actionable

MDP Component	Definition	Justification
Reward (R)	$R = w_1 \cdot (\Delta \text{ health score}) - w_2 \cdot (\text{side-effect cost}) - w_3 \cdot (\text{non-adherence penalty})$; e.g. +10 for moving to a healthier state, -5 for no change, -10 for worsening	Ties the reward signal to measurable clinical improvement, discouraging the agent from favouring actions that look good short-term but harm the patient
Transition Dynamics $P(s' s,a)$	Stochastic — e.g., from At-Risk, Medication leads to Stable with $p \approx 0.7$, stays At-Risk with $p \approx 0.2$, drops to Critical (side-effects/non-response) with $p \approx 0.1$	Real patients respond to treatment with uncertainty (adherence, physiology), so a stochastic transition model is more realistic than a deterministic one

(b) Q-Learning Update

The agent was trained over 5+ simulated patient episodes using the Q-learning update rule, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

Q-table updates for three state-action pairs across two training iterations:

Iteration	State (s)	Action (a)	Reward (r)	Next State (s')	Q(s,a) before	Q(s,a) after
1	At-Risk	Medication	+8	Stable	0.00	0.80
1	Stable	Monitor	+10	Healthy	0.00	1.00
1	Critical	Medication	-2	At-Risk	0.00	-0.20
2	At-Risk	Medication	+8	Stable	0.80	1.61
2	Stable	Monitor	+10	Healthy	1.00	1.90
2	Critical	Medication	-2	At-Risk	-0.20	-0.24

Sample calculation (Iteration 2, At-Risk/Medication): $Q = 0.80 + 0.1 \times [8 + 0.9 \times \max(Q(\text{Stable}, \cdot) = 1.00) - 0.80] = 0.80 + 0.1 \times 8.10 = 1.61$.

Convergence criterion: Training stops when the maximum absolute change across all Q-table entries between successive episodes falls below a small threshold ($\Delta Q_{\text{max}} < 0.01$), or after a fixed cap of episodes (e.g., 500–1000) with an ϵ -greedy exploration rate decayed toward 0 — whichever comes first.

(c) Final Learned Policy & Ethical Considerations

State	Greedy (Highest-Q) Action Learned
Critical	Medication
At-Risk	Medication
Stable	Monitor
Healthy	Monitor

Ethical considerations of deploying this RL agent in a clinical setting:

- Patient safety — during exploration (ϵ -greedy), the agent can try sub-optimal or unsafe actions; deployment must restrict the action space to clinically vetted options and keep a clinician in the loop for final approval, never letting the agent act fully autonomously.

- Bias — if historical treatment data under-represents certain age groups, genders, or socio-economic backgrounds, the learned policy can inherit and amplify that bias, recommending less effective care to those groups; regular fairness audits across patient subgroups are required.
- Explainability — Q-values are not inherently interpretable to clinicians or patients; using a small, auditable table-based Q-learning model (as here, rather than deep RL) and reporting the reasoning behind each recommendation supports clinical trust and informed consent.
- Accountability — responsibility for a harmful recommendation must be clearly defined; the agent should be positioned as a decision-support tool that augments the clinician's judgement, not a replacement for it.
- Reward hacking — an imperfectly specified reward function could lead the agent to optimise a proxy (e.g., minimising monitoring visits) at the expense of true patient welfare; the reward function must be validated and iteratively reviewed by medical experts before and during deployment.